Tribhuvan University
Faculty of Humanities and Social Science

A PROJECT
ON
Online Job Portal

Submitted to
Department of Humanities and Social Science
Madan Bhandari Memorial College

In partial fulfillment of the requirements for the Bachelor's Degree in Computer Application

Submitted By:

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Under the supervision of:

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Supervisor's Recommendation

This is to recommend that the project entitled "**ONLINE JOB PORTAL**", submitted by SANDEEP CHAUDHARY and SUSHANK REGMI in partial fulfillment of the requirements for the degree of Bachelor in Computer Application (BCA), has been carried out under my supervision.

…….
Signature
Sachit Ananda Mishra
PROJECT SUPERVISOR

Tribhuvan University
Faculty of Humanities and Social Science

Letter of Approval

This is to certify that this project Prepared by **Sandeep Chaudhary** and **Sushank Regmi** entitled **"ONLINE JOB PORTAL"** in partial fulfillment of the requirement of the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as project for the required degree.

Signature of Supervisor …… Sachit Ananda Mishra	Signature of HOD/Coordinator ……. Nawaraj Joshi
Signature of Internal Examiner ……. Internal Examiner	Signature of External Examiner …… External Examiner

ABSTRACT

The world is changing every second thanks to the enormous growth of Information and Communications Technology. An online job portal system aims to develop a state-of-threat platform to track employees and maintain helpful histories. The goal of the Online Job Portal System is to mechanize the prevailing manual processes using computerized equipment and developed computer applications to fulfill the needs of users. This will ensure that valuable data and information can be stored for an extended period, allowing for simple retrieval and handling. The necessary software and hardware are easily obtainable and user-friendly. The development process will be iterative, making the system user-friendly and efficient. The Online Job Portal, as described above, will be error-free, secure, reliable, and fast for job search and recruitment. It will assist users in focusing on their core activities rather than record-keeping. Consequently, it will facilitate organizations in the better deployment of resources. The system will maintain computerized records without redundant entries, ensuring users are not distracted by irrelevant data while easily accessing necessary information.

Keywords: Online Job Portal, Deployment, Computerized

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1.1 Introduction

Choosing the right career could be difficult in the fast-paced world of the future. A job portal in the future will be an online platform that acts as a central point of contact for companies and job searchers. It will serve as a marketplace where employers may publish open positions, and job seekers can go through these postings, apply, and possibly find work.

Job portals will offer a simple and efficient way to connect employers and job seekers, regardless of background or intended professional path. A job portal's future goals will mostly center on making the job search process easier for businesses and job seekers. Using this technique for online job portals will also assist employers. Employers can streamline the initial screening process, tailor questions to specific roles, and apply targeted filters using these technologies, helping narrow the candidate pool to the most qualified individuals. In general, job portals will be essential in facilitating the connection between businesses and job seekers, increasing the effectiveness and accessibility of the hiring process.

1.2. Problem Statement

A problem statement is an overview of a problem or obstacle that has to be solved. It makes the issue more understandable and acts as a roadmap for resolving it. There are various possible problem statements related to employment that companies or developers may want to take into consideration. The job platform is challenging to use, which causes dissatisfaction and lowers interest. Some job listings are confusing to applicants because they are out of current or contain important information that is missing. The employment recommendations that users obtain are broad and do not correspond with their own interests or career objectives. There are less job ads on the platform as a result of employer's inactive participation. The uploaded resumes are difficult for the system to process and extract suitable information.

1.3. Objectives

The objective of an online job portal is to facilitate connections and meet the needs of businesses and job seekers by offering a platform.

i. Job listing and finding jobs:

To list and find the job.

ii. **Easy profile customization:**

To customize their profiles by giving them simple tools to alter their preferences, experiences, and talents.

iii. **Easy Job Applications:**

Users can apply to jobs quickly without filling out long, complicated forms each time.

1.4. Scope and Limitation

Every website has its own unique features and its limitations. The website offers following scope and lacks following things

1.4.1. Scope

1) Users can log in and find the job available on a dashboard.
2) Users with no experience can easily browse through the website.
3) User friendly UI.
4) Users can get notification on applied jobs.

1.4.2. Limitation

The different limitations of this project are:

- Since the system is web based, it might not be accessible to people not having access to an internet connection.
- This website has limited features for the registered users that means only the registered users can see the job available and enter the job.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1. Background Study

In today's world the use and access to the internet is so high that most people are busy with their own work so we have developed this module so that users can create and access their account using internet and general concept and terminologies are mentioned below:

a. **Create New Account**: A user can create an account through registration process and user can create an account when there is access to the internet through this module.
b. **Login**: After creation of account users can login through their details and can access their job portal through this module,
c. **Browsing**: After successful login, users can browse through different categories as admin, job seeker or employee.
d. **Apply Jobs**: Job seekers can conveniently submit applications for positions that align with their skills and qualifications.

2.2 Literature Review

Traditional job-seeking methods are often slow, stressful, and challenging, lacking in quality and efficiency. Job seekers have to invest a significant amount of time and effort in gathering information, preparing applications, and considering costs associated with the process. However, the emergence of online job portals has revolutionized the job seeking landscape, providing a faster and more convenient alternative. The Internet has become a powerful tool for job seekers, offering numerous websites that advertise job vacancies across various industries and skill sets. This shift towards online platforms has significantly improved access to job opportunities. Moreover, the Internet plays a crucial role in human resource planning and development, with many organizations utilizing computer technology and the Internet for recruitment purposes. It is important to note that while the Internet has streamlined the job-seeking process, it has not completely replaced traditional methods. Offline methods, such as networking, referrals, and direct application submissions, still hold value in certain scenarios. However, the convenience and accessibility provided by online job portals have made them a primary resource for job seekers in today's digital era.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1. System Analysis

Considering the fact that this project involves design and implementation of a software system regardless of that is web-based, it will be important to mention and consider some models used in software development and deployment, some general models of software development are namely:

i. **The Waterfall approach**: It represents activities in requirements, specifications, design, implementation and testing. All these as separate processes.
ii. **Incremental Evolutionary development**: It involves rapid development of the specifications and then refined later for the customer.
iii. **Formal transformation**: This approach is based on setting and producing a mathematical specification for the system to meet.
iv. **System assembly from reusable components**: This approach assumes that part of the system already exists. This model is focused on integration

After reading though all these models. The waterfall model fits the development of this website. The main aim of using this approach is we can focus on each part of the model during development and come back to it if need be. The project can easily break down into different parts based on this model.

This is the model that will be used to develop the E-Commerce system. However, feedback loops will be allowed during the whole software development process. The model chosen for this project has to favor two developers for a project. Because we are the only ones who are going to implement this project. We find this model suitable for us to follow.

It requires that software development follows the following stages:

a. Requirements are to be proposed
b. System design should be made according to the requirements
c. Implementation of the features according to the design
d. Integration and testing of the system
e. Deployment of the system
f. Maintenance of the system

This process has been illustrated below on Figure showing top-down development.

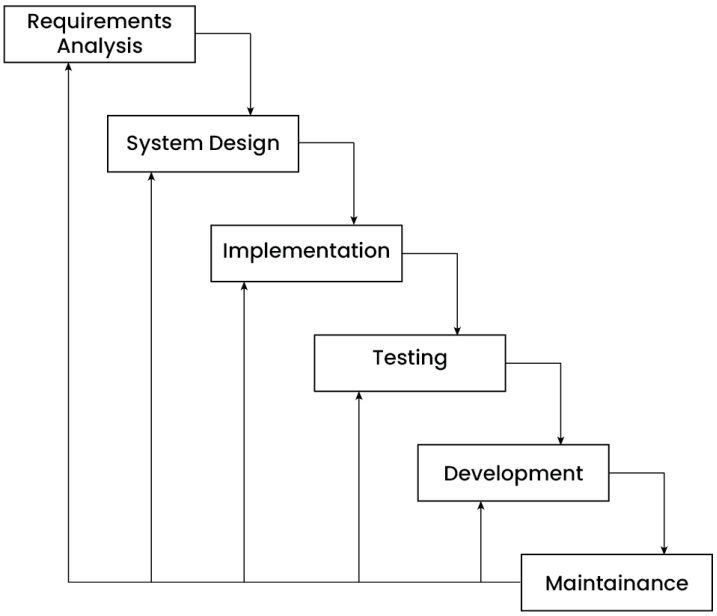

Figure 1: Waterfall model

3.1.1 Requirement Analysis

Requirement analysis is done while developing a system and before implementing it, it is necessary to analyze the whole system requirement. It is category into mainly two parts:

i. Functional requirements

ii. Non-functional requirements

For any system, there are functional and non-functional requirements to be considered while determining the requirements of the system. The functional requirements are user "visible" features that are typically initiated by stakeholders of the system, such as generate report, login, and signup. On the other hand, non-functional requirements are requirements that describe how the system will do what it is supposed to do, for example, Usability, Reliability & Availability, Performance, Security and maintainability.

i. Functional requirement: The requirement that has been used in the project as the functional requirements generally include functions such as input, the processing and the final output. The functional requirements in the project are mentioned below:

1. User Module

i. Users can register and log in to the system.

ii. Users can view updated job details after modification.
iii. Users can delete or cancel jobs if required.
iv. Users can log out from the system after completing tasks.
v. Users can apply for the jobs from listed jobs.

2. Admin Module

i. Admin can login to the system.
ii. Admin can see the registered users.
iii. Admin has privilege to delete the user.
iv. Admin can log out from the system.

ii. Non-Functional Requirement: The non-functional requirement specifies how the system works. The non-functional requirements included in the project are:

a. System uses different databases for storing the attribute for each entity.
b. Users receive notifications on their job applications.
c. Users can see the job details after entering the job available or vacancy.

3.1.2. Feasibility Analysis

i. Technical Feasibility

a. The UI of our project is very simple.
b. Users will require internet browser and internet to use it.
c. It will run on many existing web browsers with the latest versions and even in smart phones.

ii. Operational Feasibility

These include reliability, maintainability, usability, supportability. The proposed system is operationally feasible as it is reliable for all types of users, i.e. whether or not. The user has a knowledge of computers or not. The proposed system is supported by a small to large-scale organization. It is simple and easy to use due to simple user interface and its operational feasibility.

iii. Economic Feasibility

Before the development of a system, the proposed system should be studied whether it is within the budget estimated by the organization. The project that we are developing is within the cost estimation of the organization. The project cost is less, and no more burdens are

needed. The system development does not have any requirement for expensive hardware and software. The platforms are open sources and the resources required for the project are also open source. Hence the project is said to be economically feasibility.

iv. Schedule Feasibility

The project expands over a period of 120 days with Task being divided along 6 with duration in the time frame. For the system to be delivered in the specified time frame we have to make sure deadlines are met and tasks are executed accordingly in the specified duration period.

Table 1: Gantt Chart

Time	Day 1 - 15	Day 16 - 45	Day 46 – 75	Day 76 - 105	Day 106 - 115	Day 115 - 120
Planning						
Research and analysis						
Design						
Implementation						
Testing						
Documentation						

3.1.3 Use Case Diagram

A Use Case Diagram is a vital tool in system design; it provides a visual representation of how users interact with a system.

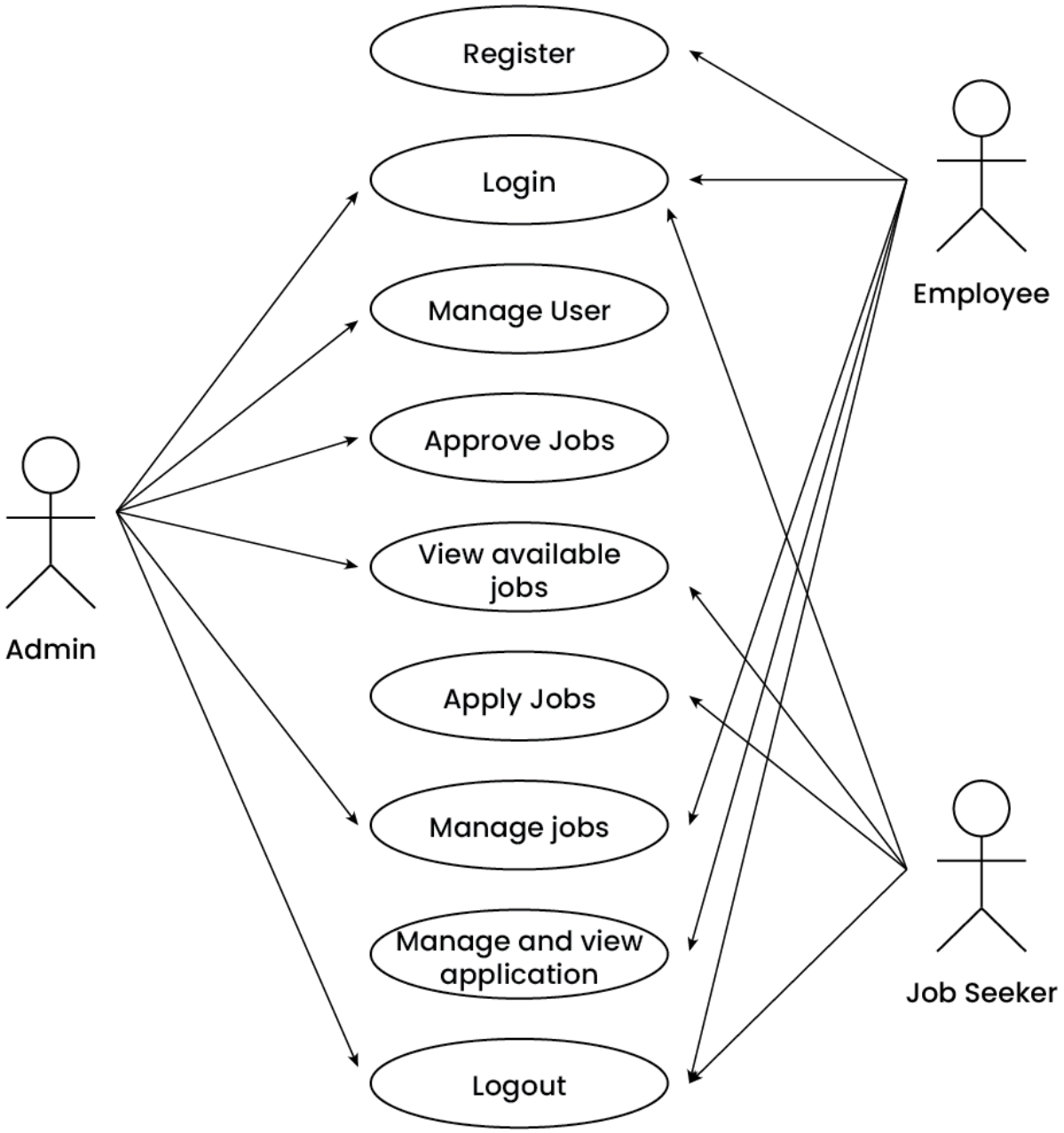

Figure 2: Use case diagram of job-portal

3.1.4 Data Modeling (ER-Diagram)

In our database system user and account are the entities which contain a primary key as a unique identifier for each entity and other attribute to show the properties of these entities. The overall relationship between the entities is shown in the ER diagram below

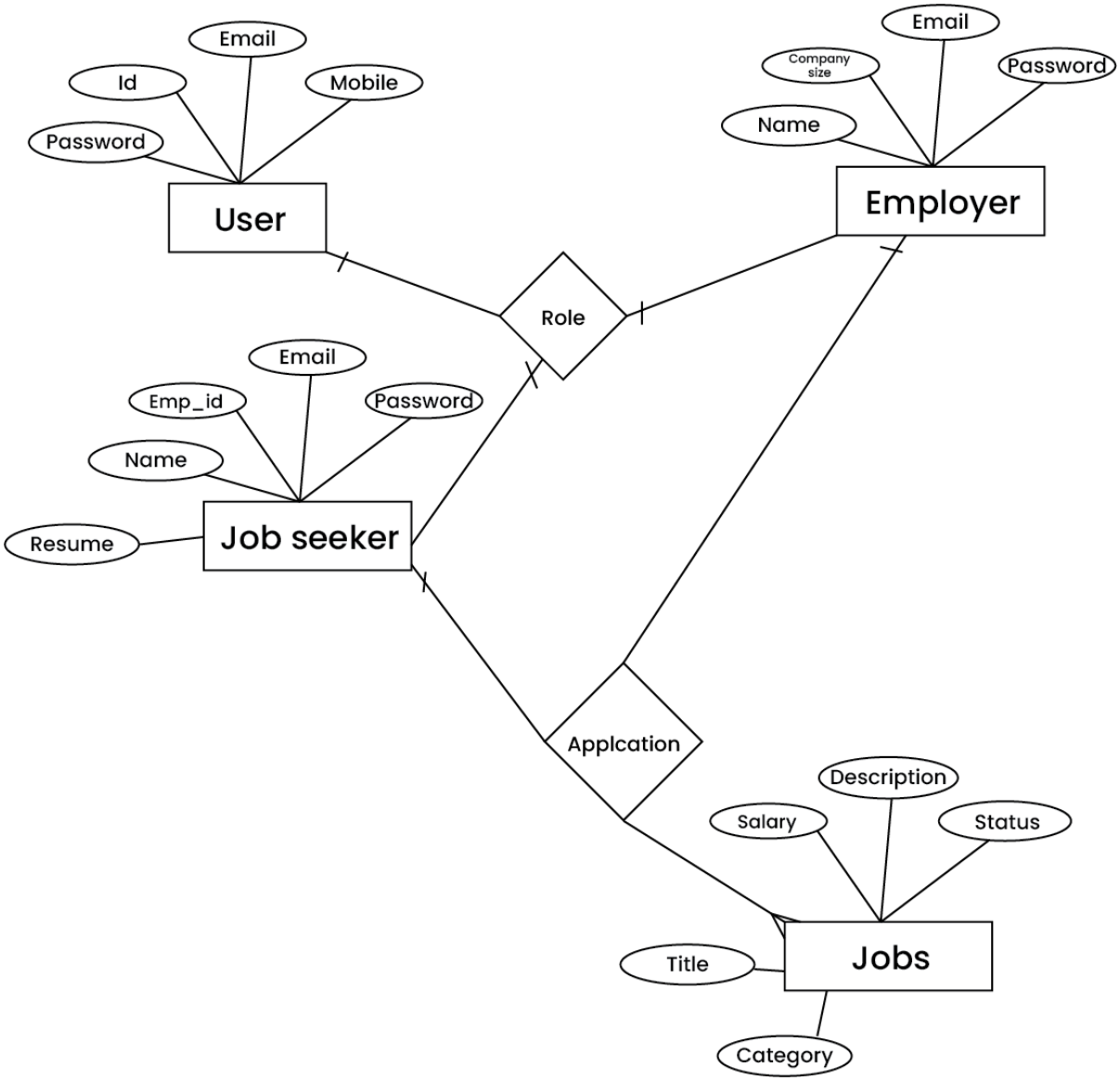

Figure 3: E-R diagram of the job-portal

3.1.5 Process Modeling (DFD)

Process modeling is the technique of graphically capturing the series of processes and showing how they are related to one another.

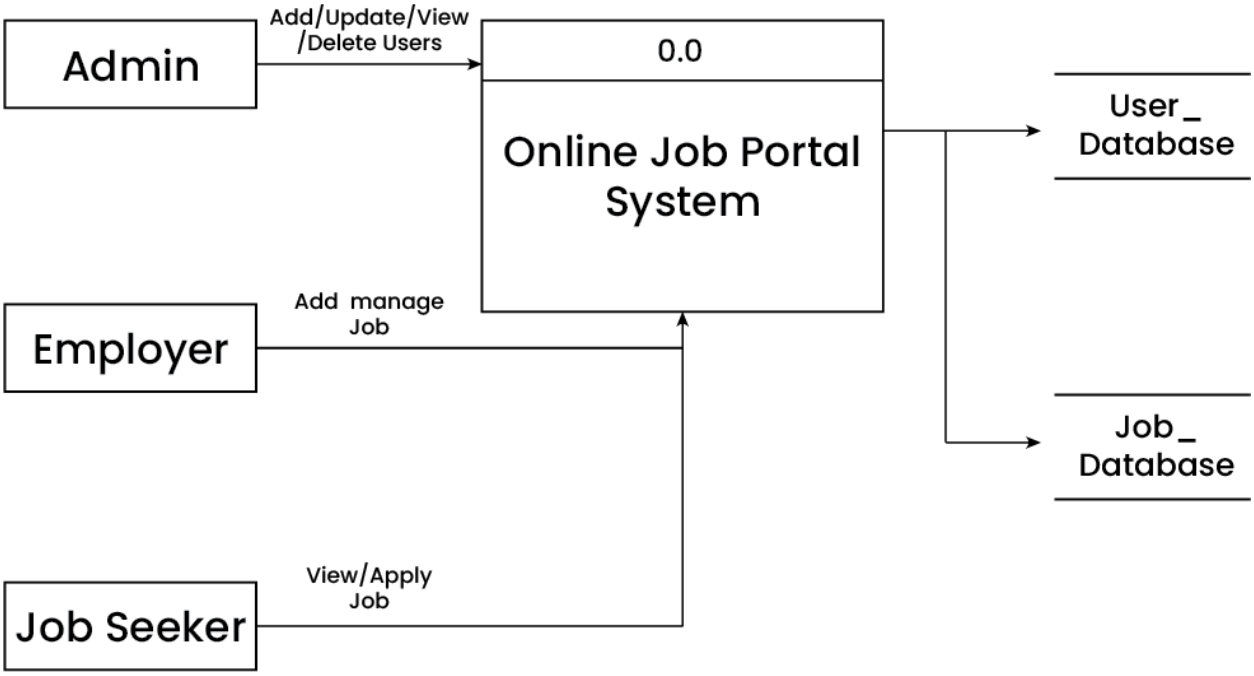

Figure 4: Dfd level 0 (Job-portal)

This Level 0 Data Flow Diagram (Context Diagram) illustrates the Online Job Portal System, showing how the Admin manages user data (adding, updating, viewing, deleting users) stored in the user_database, how Employees add and manage 9 job listings stored in the job_database, and how Job Seekers view and apply for jobs within the system.

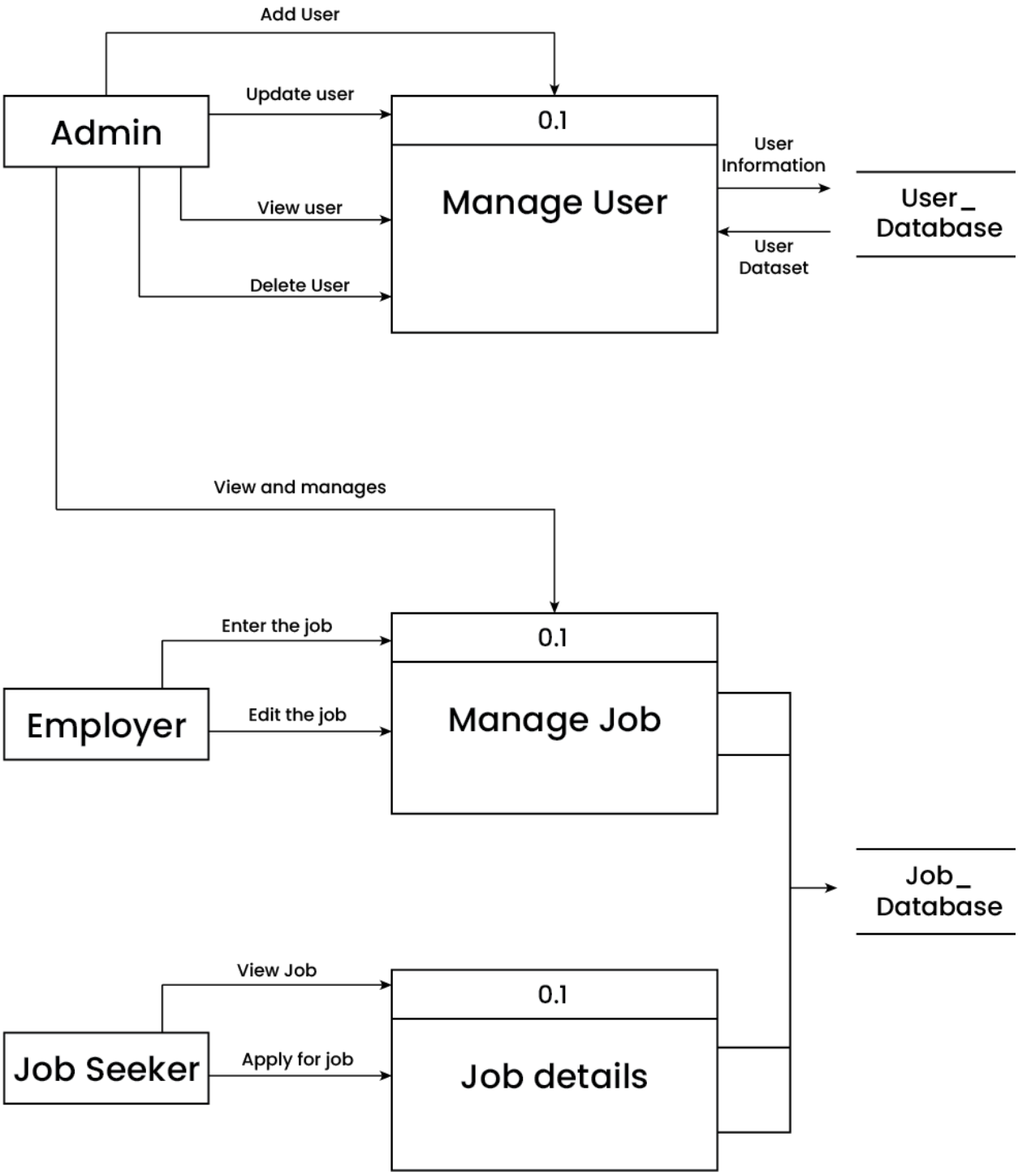

Figure 5: Dfd level 1 (Job-portal)

The provided flowchart illustrates the hierarchical structure and functional processes of a user management and job application system, involving three main actors (Admin, Employee, and Job Seeker) interacting with specific databases ('user database' and 'job database'). This diagram can be relevant for understanding user roles and permissions in the system.

3.2. System Design

3.2.1 Architecture Design

In the job portal website, users interact with the system through a simple user interface. The job portal website uses three tier architecture. The data is collected from the users and stored in the database through which the server provides detail to the user. To perform transactions to the user, the system uses different database tables including different attributes for each entity, user has a unique key which makes them different from other users. In this way-out system architecture is designed which is an abstract view of the system.

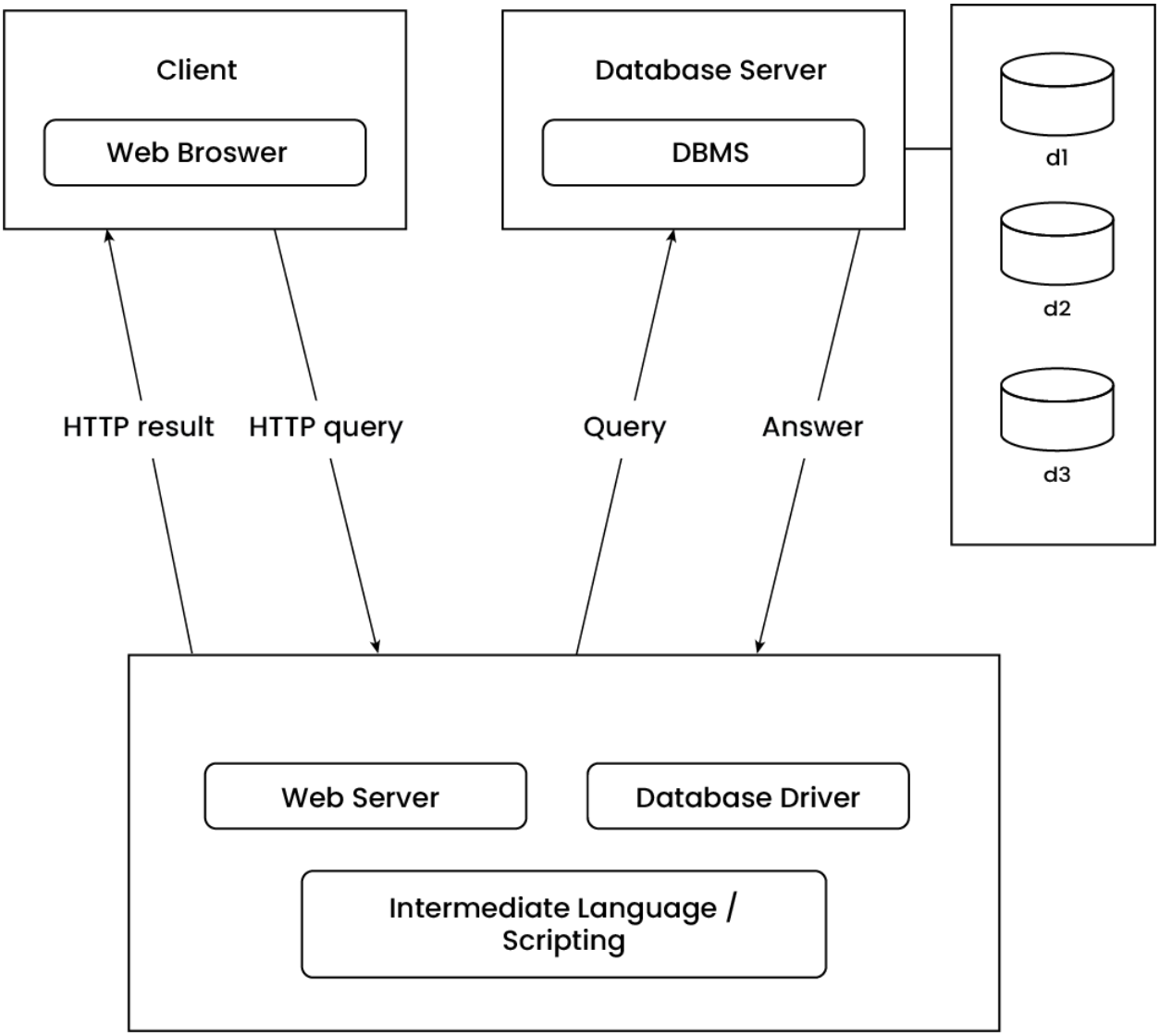

Figure 6: System architecture

3.2.2 Database Schema Design

The database schema design for job portal showing all the relations along with their attributes and inter-relationship between the relations is shown below:

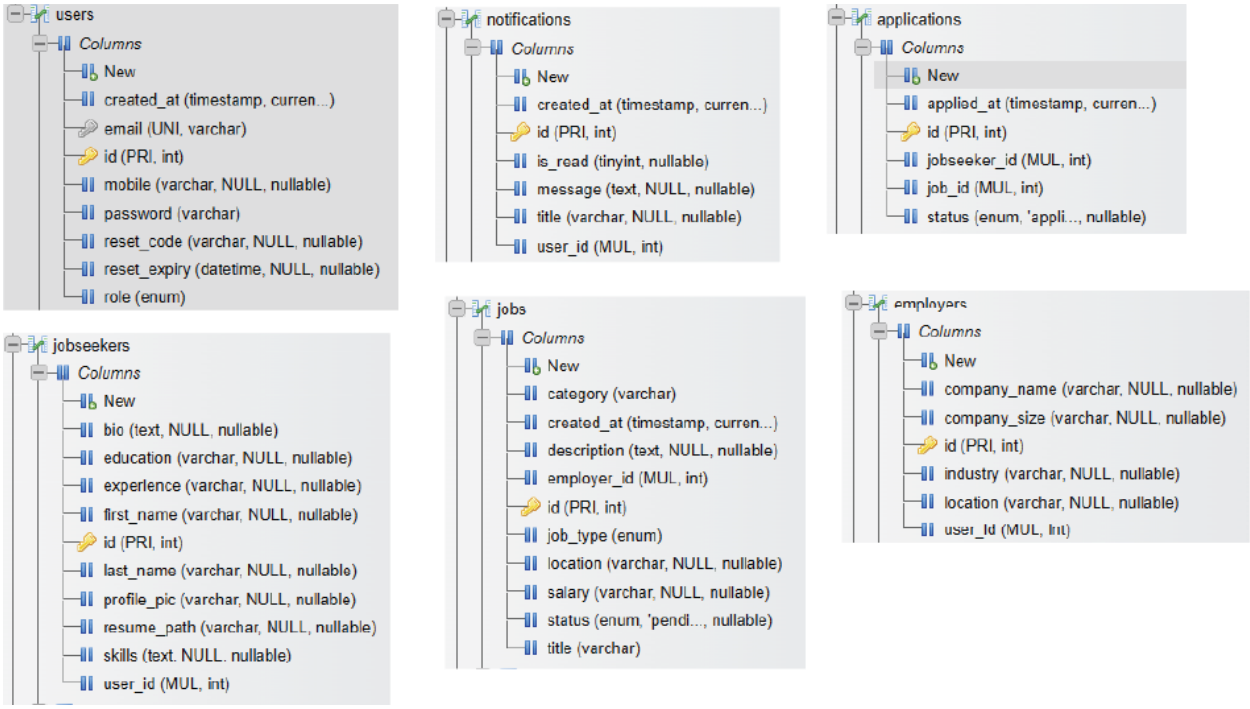

Figure 7: database schema design

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

The implementation phase of the Online Job Portal involves converting the system design into a functional application using appropriate technologies and tools. The system was developed following a modular approach, ensuring scalability, maintainability, and ease of future enhancement. The frontend handles user interaction and interface rendering, while the backend manages business logic, authentication, and database operations. The system enables job seekers to search for and apply for jobs and allows employers to post and manage job vacancies efficiently.

4.1.1 Tools Used

The following tools and technologies were used for the development of the Online Job Portal:

- **Frontend:** HTML, CSS, JS
- **Backend:** PHP
- **Database:** MySQL
- **Code Editor:** Visual Studio Code
- **Version Control:** Git
- **Web Browser for Testing:** Google Chrome

These tools were selected due to their flexibility, performance, and ease of integration for web-based applications.

4.1.2 Implementation Details of Modules

The system was divided into several functional modules:

1. User Module (Job Seeker)

This module allows job seekers to:

- Register and log in securely
- Create and update personal profiles
- Finding available jobs
- Apply for available jobs
- Update about the jobs applied

2. Employer Module

The employer module enables recruiters to:

- Register and manage company accounts
- Post new job vacancies
- Edit or delete job listings
- View and manage applicants
- Shortlist candidates based on requirements

3. Admin Module

The admin module provides full control over the system:

- Manage users and employers
- Approve or remove job postings
- Monitor overall system activity
- Maintain database integrity and security

4. Application Approval

This module provides dynamic job Approval based on:

- Rejecting applicants
- Interview applicant
- Hired applicant

5. Application Management Module

This module handles:

- Job application submission
- Application status tracking
- Notification to employers about new applicants

4.2 Testing

Testing was performed to ensure that the system works correctly, meets requirements, and is free from major errors. Different testing techniques were applied to verify functionality, usability, and reliability of the job portal.

4.2.1 Test Case for Unit Testing

Test Case	Module	Input	Expected Output	Result
TC 01	User Registration	Valid User Details	Account Created Successfully	Pass
TC 02	User Login	Correct email and password	Login Successful	Pass
TC 03	Job Post	Enter job vacancy details	Job vacancy posted	Pass
TC 04	Job Approve	Posted job approved by admin	Application Approved	Pass

TC 05	Application approval	Click approve buttons (Interview/Reject/hired)	Select as (Interview/Reject/Hired)	Pass

Unit testing ensured that each module performs its intended function independently. Integration testing was also performed to verify the interaction between frontend, backend, and database.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Lesson Learnt / Outcome

During the development of the Online Job Portal, several important technical and practical lessons were learned:

- Understanding of full-stack web development process
- Experience in designing user-friendly interfaces
- Knowledge of database schema design and API integration
- Improved problem-solving and debugging skills
- Importance of system testing and validation
- Experience working with modular architecture for scalability

This project enhanced both technical knowledge and project management skills.

5.2 Conclusion

The Online Job Portal was successfully designed and implemented to bridge the gap between job seekers and employers through a centralized web platform. The system allows job seekers to easily search and apply for jobs, while employers can efficiently post vacancies and manage applications. The portal provides a simple, responsive, and user-friendly interface supported by a secure backend and well-structured database. Overall, the system fulfills the primary objective of simplifying the recruitment process and providing a digital solution for employment management.

5.3 Future Recommendations

Although the current system meets its core objectives, several improvements can be made in the future:

- Integration of AI-based job recommendation system

- Resume parsing and automatic skill extraction
- Real-time chat system between employer and job seeker
- Email and SMS notifications for job updates
- Mobile application version of the portal
- Advanced analytics dashboard for employers
- Integration with third-party professional platforms

These improvements will enhance usability, automation, and user engagement, making the Online Job Portal more efficient and competitive.

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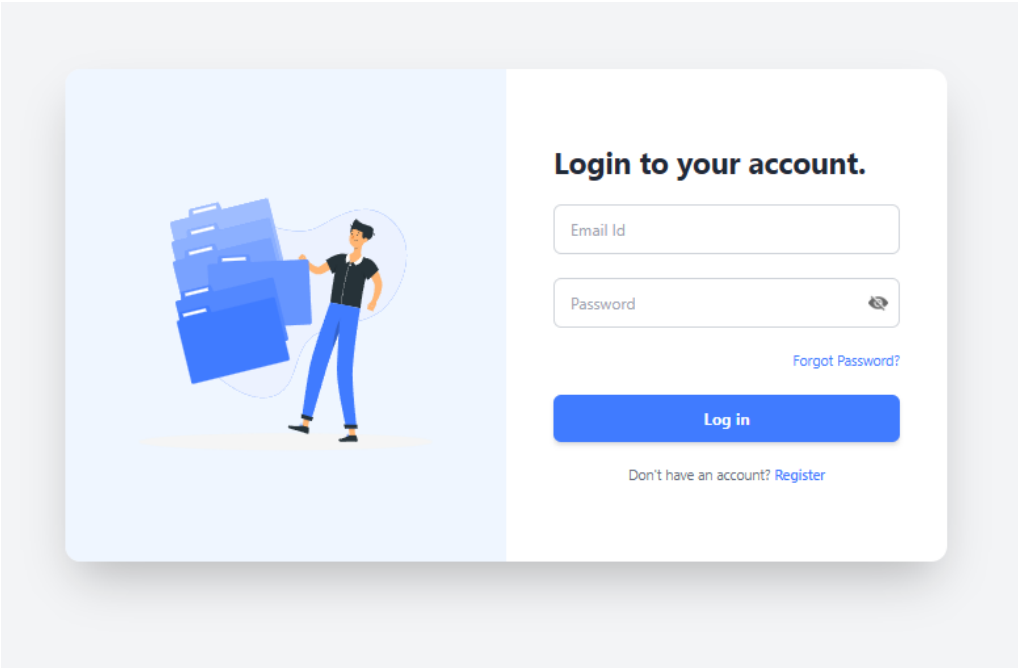
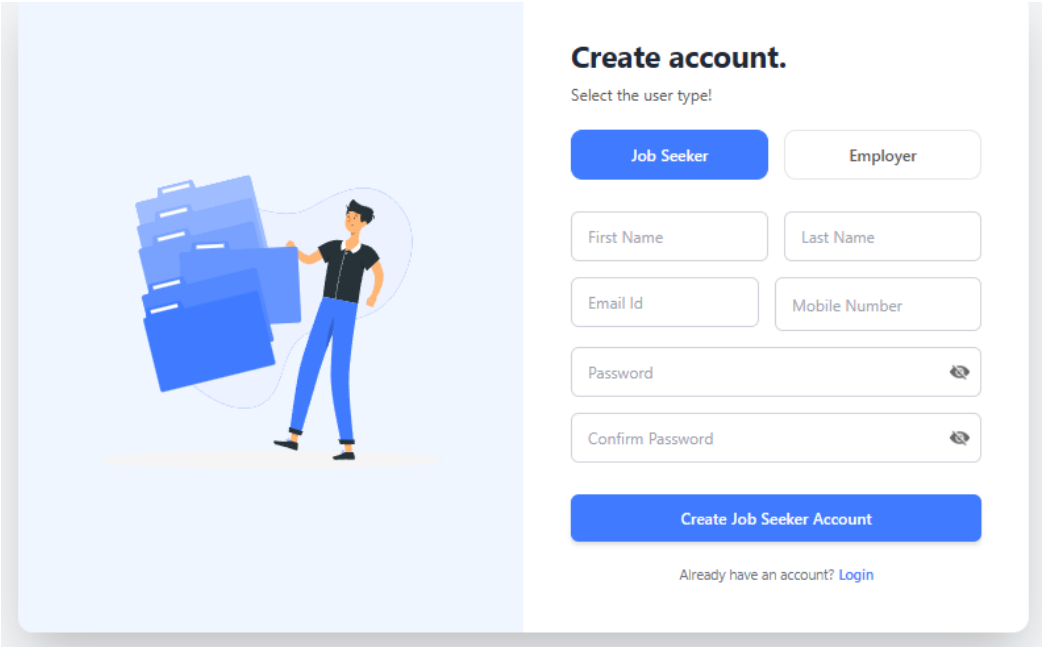
[3] Tutorial Republic, "PHP MySQL CRUD Application." Available: https://www.tutorialrepublic.com/

Appendices

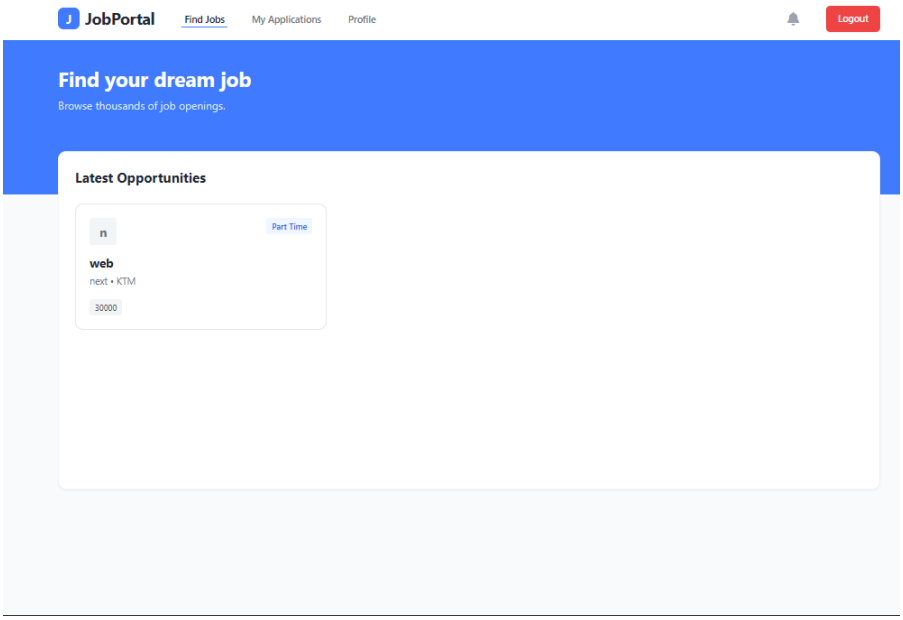
1. Login Page



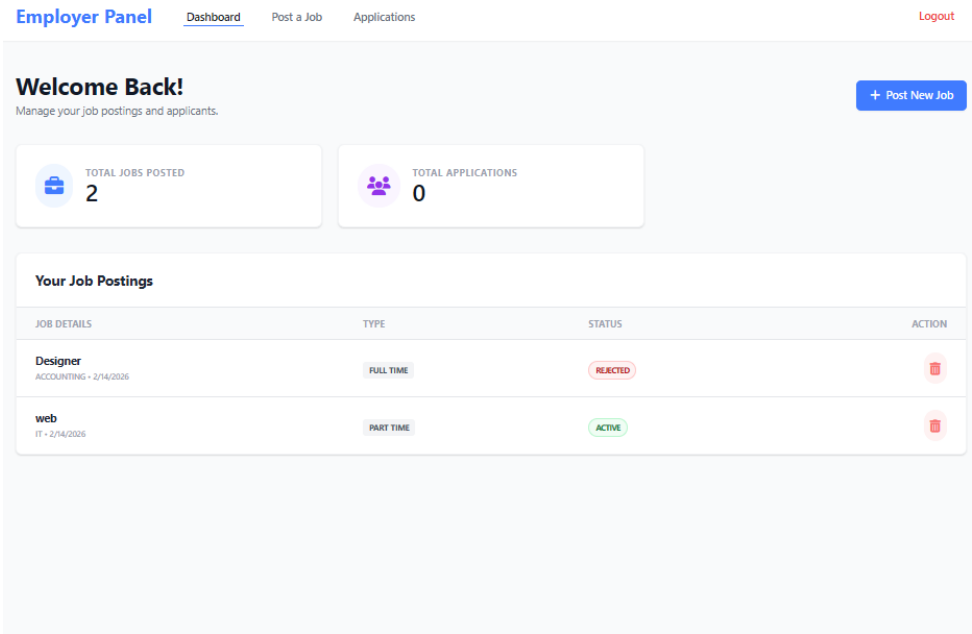
2. Job Seeker registration



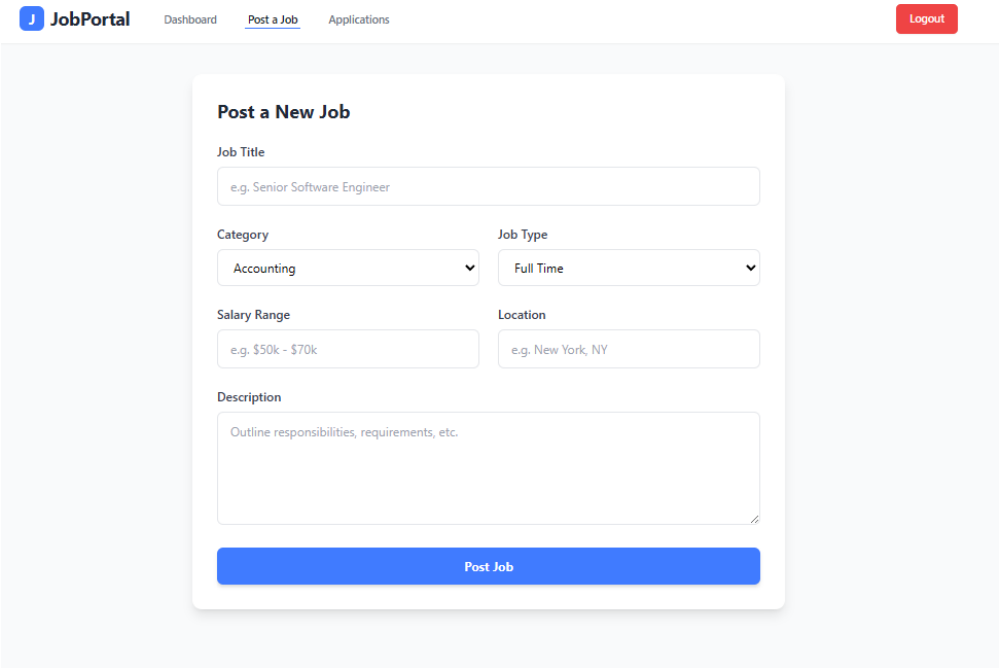
3. Job Seeker dashboard



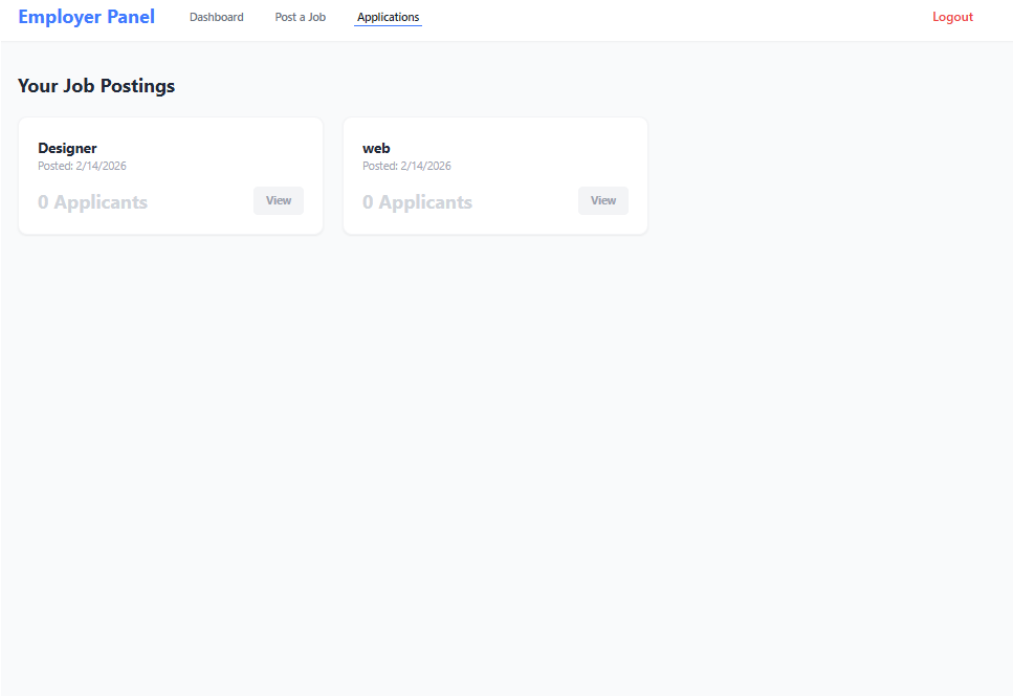
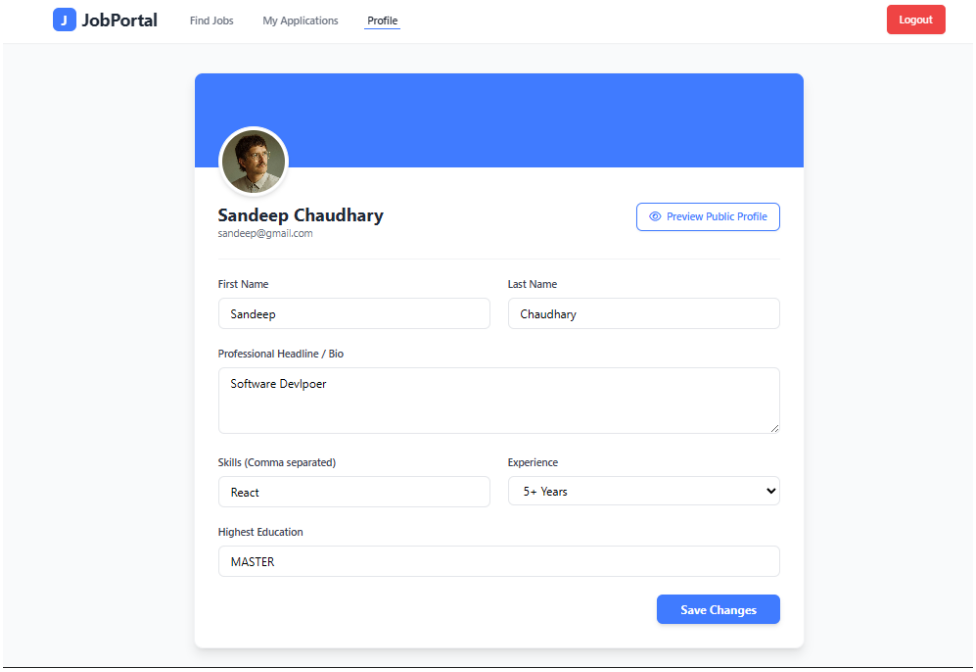
4. Employer Dashboard



5. Employer Post a job



6. Employer Applications



7. User Profile